

KEN CHENG

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EDUCATION

Columbia University

B.S. in Computer Science; Statistics Minor; GPA: 3.8

Expected May 2028

Egleston Scholar (Top 1%)

- Coursework: Machine Learning, Natural Language Processing, Probability Theory, Linear Algebra, Databases

EXPERIENCE

Columbia University | Creative Machines Lab

September 2026 – Present

Undergraduate Deep Learning Researcher

PyTorch | Hugging Face Transformers | scikit-learn

- Trained **3.18M-parameter transformers** and MLPs to study deceptive communication; ran an **18-job ablation across three seeds** comparing output-head designs and factual retention under conflicting rewards
- Implemented layer/attention-head activation extraction, training-only PCA, model-specific adapters, and a shared neural detector; added tests for data leakage, serialized predictions, and frozen-classifier adaptation

Amazon

June 2026 – August 2026

Software Development Engineer Intern

Python | AWS Lambda | S3 | AWS CDK (TypeScript)

- Designed and shipped long-term forecasting configuration infrastructure across **17 packages and services**, used by **30+ engineers, managers, economists, and scientists**; owned testing, rollout, and documentation
- Automated supply-chain model dependency tracing, reducing manual comparison time by **~90% based on user surveys**; delivered forecasting configurations with AWS Lambda, S3, and AWS CDK

Quantiv

July 2025 – Present

Founder & Lead Engineer | usequantiv.com

Python | LightGBM | DuckDB | Parquet

- Trained **six LightGBM models** for an open-source earnings forecasting platform, achieving **31% lower weighted MAE** than the options-implied baseline on development validation; produced point and quantile predictions
- Built feature and batch-inference pipelines with DuckDB/Parquet and **four-fold walk-forward validation** using a five-day purge; delivered forecasts to **120+ monthly active users** with data-freshness and drift monitoring
- Hardened FastAPI model serving with signed artifacts, SHA-256 verification, and atomic model activation

RESEARCH & PROJECTS

GC-INF: Graph Neural Network & Transformer Forecasting | [Paper](#)

September 2022 – October 2023

Sole-Author Paper, IEEE Intelligent Transportation Systems Conference (2023)

PyTorch | GNNs

- Developed a graph-convolutional and Informer model; evaluated **four baselines across three horizons**, reducing 15-minute turning-ratio RMSE by **24%** versus STGCN on SUMO-simulated traffic data

Financial Document Retrieval Engine (FDRE) | thefdre.com

June 2026 – Present

Open-source Hybrid Retrieval & Citation-Verified Answers

Python | FastAPI | PostgreSQL | pgvector

- Built hybrid lexical/vector retrieval over **3.04M embedded passages from 3,204 SEC filings**, serving **75+ monthly active users** with citation checks, evidence-based abstention, and point-in-time filtering
- Evaluated multi-query expansion in a **33-query ablation**, improving Recall@5 from **0.152 to 0.212**; reduced database storage **27% (15 to 11 GB)** with half-precision vectors

Wasserstein Regimes | [GitHub](#)

September 2026 – Present

Open-source Optimal Transport & Unsupervised Learning

Python | NumPy | scikit-learn

- Compared **nine clustering methods across five ETFs** using chronological validation, purged holdouts, and synthetic controls; isolated volatility effects from distribution-shape effects in market-regime clustering

AWARDS

USA Computing Olympiad (USACO): Platinum (highest division); 1000/1000 Gold promotion score (2023)

National Winner, High School Big Data & AI Challenge – STEM Fellowship (2024)

Co-authored carbon-sequestration research analyzing LiDAR canopy heights and basal-area rasters in Python and Rasterio.

Canada-Wide Science Fair Gold (2022); **LA Hacks 2025**: Top 5/172 teams, Google Gemini and Financial Tech prizes

TECHNICAL SKILLS

Languages / Data: Python, C++, SQL, Bash, NumPy, Pandas, SciPy, PostgreSQL, pgvector, DuckDB, Parquet, Apache Spark

Machine Learning: PyTorch, JAX, Hugging Face Transformers, scikit-learn, LightGBM, CUDA, GPU Profiling, MLflow

LLMs / Agents: Retrieval-Augmented Generation (RAG), LLM Evaluation, Inference Optimization, Tool Calling

Cloud / Tools: Amazon Web Services (AWS): EC2, S3, Lambda, DynamoDB, VPC; Docker, CI/CD, Git, Linux